

MEDICAL IMAGE PREPROCESSOR

Complete AI Agent Implementation Guide

Modality-Aware Preprocessing Pipeline for Medical LLM Input

Version 1.0 | For VS Code AI Agent Execution

0. Purpose & Guiding Principles

This document is a complete, step-by-step implementation guide for an AI coding agent operating inside VS Code. It describes what to build, why each component exists medically, and exactly how each module should behave. No code is provided intentionally — the agent must implement all logic from these descriptions.

0.1 What This System Does

The Medical Image Preprocessor accepts raw medical images in DICOM, PNG, JPEG, or TIFF format, automatically identifies the imaging modality, applies a sequence of medically appropriate preprocessing steps, and outputs a clean, standardized image ready for ingestion by a downstream medical large language model (LLM) for findings extraction.

The pipeline must never alter, destroy, or misrepresent diagnostic information. Every transformation must be reversible in principle, and every step must be justified by an improvement to signal quality for either human radiologists or the downstream LLM.

0.2 Why Preprocessing Matters for the Medical LLM

Medical LLMs that extract findings from images — for example, identifying pneumothorax on a chest X-ray, white matter lesions on MRI, or nodular densities on CT — perform significantly better when given preprocessed input because:

- Standardized resolution and orientation remove the burden of geometric normalization from the model.
- Proper contrast enhancement makes subtle findings (small nodules, early infiltrates, hairline fractures) more salient without fabricating structures.
- Noise reduction improves edge clarity so the LLM's visual encoder can distinguish real anatomy from acquisition artifact.
- Consistent bit-depth and dynamic range prevent attention collapse on extremely bright or dark regions.
- Modality-specific windowing (for CT) ensures the LLM sees the same tissue contrast a radiologist would see on a calibrated workstation.



CRITICAL MEDICAL CONSTRAINT

Preprocessing must NEVER introduce, remove, or modify clinically significant structures.

Do not apply lossy compression to DICOM outputs. Preserve the original file alongside any processed version.

Never apply color jitter, hue shift, or saturation changes to CT, MRI, or PET — pixel values in these modalities carry quantitative clinical meaning (Hounsfield Units, T1/T2 signal intensity, SUV).

Histopathology and dermoscopy are the ONLY modalities where color transformation is clinically appropriate.

All quality metrics must be logged. If quality drops after any step, that step must be flagged and optionally skipped.

1. Project Structure

Create the following directory layout at the root of the project. Every module described in this guide maps to one of these files. The agent should create each file as an empty stub first, then implement them in the order given in this guide.

File / Directory	Purpose
pipeline.py	Orchestrator — the single entry point that chains all stages in order
config/modality_configs.yaml	Per-modality default parameters (sigma values, CLAHE clip limits, window presets, etc.)
core/dicom_handler.py	DICOM I/O, pixel array extraction, metadata parsing
core/modality_detector.py	Classifies the imaging modality from DICOM tags or ML fallback
core/standardizer.py	Resize, bit-depth normalization, orientation correction
core/quality_assessor.py	Pre- and post-processing quality metrics (SNR, CNR, sharpness)
core/llm_formatter.py	Final output formatting and metadata packaging for the medical LLM
filters/base_filter.py	Abstract base class all modality filters must inherit from
filters/ct_filters.py	CT-specific noise reduction, windowing, artifact handling
filters/mri_filters.py	MRI-specific bias field correction, NLM denoising, skull stripping
filters/xray_filters.py	X-ray specific CLAHE, Wiener denoising, beam-hardening correction
filters/ultrasound_filters.py	Ultrasound speckle reduction via anisotropic diffusion
filters/histopath_filters.py	Stain normalization, color deconvolution, tile management
filters/fundus_filters.py	Retinal image CLAHE on green channel, illumination normalization
filters/dermoscopy_filters.py	Hair artifact removal, vignette correction, lesion isolation
agents/filter_selector.py	Sub-agentic router that selects optimal filter config per image
utils/visualization.py	Side-by-side before/after display utilities for debugging
utils/logger.py	Structured logging with step-level timing and quality delta tracking
tests/test_pipeline.py	Integration tests against one sample image per modality



REQUIRED LIBRARIES — Install These First

pydicom — DICOM file reading and metadata extraction

SimpleITK — 3D volume I/O, resampling, N4 bias field correction for MRI

opencv-python (cv2) — General image operations, CLAHE, morphological ops

scikit-image (skimage) — NLM denoising, Wiener filter, quality metrics

scipy — Signal processing utilities (Wiener, Gaussian)

numpy — All array operations

monai — Medical imaging augmentation and transforms (optional but recommended)

staintools — Macenko and Vahadane stain normalization for histopathology

medpy — Anisotropic diffusion for ultrasound speckle reduction

noisereduce — Spectral noise gate (used for SNR estimation utility)

Pillow — PNG/JPEG I/O fallback

pyyaml — Config file parsing

loguru — Structured logging

2. Configuration File (config/modality_configs.yaml)

Create a YAML configuration file that stores default parameters for every modality. The pipeline reads this file at startup. Every numerical parameter used anywhere in the pipeline must be sourced from this config — no hardcoded values in filter modules. This makes the system tunable without code changes.

2.1 What to Include Per Modality

Each modality key should contain the following sub-keys:

- `target_size`: The standard output resolution in pixels (width, height). For CT and MRI, use 512×512. For histopathology tiles, use 224×224. For chest X-ray, use 1024×1024. For ultrasound, use 512×512. For fundus, use 800×800.
- `bit_depth_output`: Either float32 (range 0.0–1.0) or uint8 (0–255) depending on downstream LLM requirements. Default to float32 for all modalities.
- `clahe_clip_limit`: Floating point. Controls how aggressively CLAHE redistributes contrast. Higher values increase contrast but risk noise amplification. Medically validated defaults: 2.0 for X-ray, 3.0 for fundus, 1.5 for dermoscopy.
- `clahe_tile_grid_size`: Tuple of (rows, cols). Smaller tiles = more local contrast. Recommended: (8,8) for most, (16,16) for high-resolution histopathology.
- `denoise_strength`: Floating point 0.0–1.0. Maps to filter-specific sigma or h parameter.
- `ct_window_presets`: Dictionary of named window/level pairs. See Section 5 for full list.
- `mri_bias_correction`: Boolean flag — whether to apply N4ITK bias field correction.
- `orientation_standard`: String indicating target anatomical orientation (e.g., 'LPS', 'RAS').
- `quality_snr_threshold`: Minimum acceptable SNR in dB before flagging the image.
- `llm_output_format`: String, either 'png' or 'jpeg'. PNG recommended to avoid compression artifacts.

3. DICOM Handler (core/dicom_handler.py)

This module is the entry point for all DICOM input. It must handle the full range of DICOM complexity encountered in real clinical datasets. Non-DICOM inputs (PNG, JPEG, TIFF) fall through to a lightweight PIL-based loader in the same module.

3.1 Loading and Pixel Array Extraction

- Use `pydicom's dcmread` function to load the DICOM file. Always pass `force=True` to handle malformed headers from older scanners.
- Check for the `PixelData` attribute before any processing. If missing, raise a descriptive error — the file may be a DICOM SR (Structured Report) or DICOM PR (Presentation State), neither of which contains image data.
- Extract the pixel array using the `pixel_array` property. This returns a NumPy array. For 16-bit CT or MRI data, the array dtype will be `int16` or `uint16`.
- Apply Rescale Slope and Rescale Intercept corrections for CT. These DICOM tags convert the stored integer values into true Hounsfield Units (HU). The formula is: $HU = pixel_value \times RescaleSlope + RescaleIntercept$. If these tags are absent, assume `slope=1` and `intercept=0`. This correction is mandatory for CT — skipping it produces wildly wrong tissue contrast.
- For multi-frame DICOM (3D CT or MRI volumes), the `pixel_array` will have shape (frames, rows, cols). Extract a representative 2D slice for LLM input. Default to the middle slice for brain MRI, and to the slice with maximum variance for chest CT (this tends to be the slice with the most diagnostic content).

3.2 Metadata Extraction

Extract and return the following DICOM tags as a structured metadata dictionary. This metadata will be passed alongside the processed image to the LLM formatter in Section 9.

- Modality (0008,0060) — e.g., CT, MR, CR, DX, US, NM, PT, MG
- BodyPartExamined (0018,0015) — e.g., CHEST, HEAD, ABDOMEN
- PatientAge (0010,1010) — anonymize or remove for non-clinical use
- PatientSex (0010,0040) — relevant for organ-specific preprocessing norms
- StudyDescription (0008,1030) and SeriesDescription (0008,103E)
- KVP (0018,0060) and Exposure (0018,1152) for X-ray — these affect noise characteristics
- MagneticFieldStrength (0018,0087) for MRI — 1.5T vs 3T affects noise floor and bias severity
- ImageOrientationPatient (0020,0037) — needed for anatomical orientation correction
- SliceThickness (0018,0050) — relevant for 3D volume handling
- PhotometricInterpretation (0028,0004) — MONOCHROME1 vs MONOCHROME2 vs RGB. MONOCHROME1 means high pixel value = black (inverted). Must be detected and corrected before any further processing.

! MONOCHROME1 INVERSION — Common Mistake

Some X-ray DICOMs use MONOCHROME1 photometric interpretation where bright pixel values represent DARK areas (bone = dark, air = bright).

This is the OPPOSITE of conventional radiographic viewing and the opposite of what most ML models expect.

If PhotometricInterpretation == 'MONOCHROME1', invert the pixel array ($\text{max_value} - \text{pixel_array}$) BEFORE any other step.

Failure to handle this will cause the LLM to interpret anatomy backwards — a critical clinical error.

3.3 Non-DICOM Fallback Loader

- If the input file has extension .png, .jpg, .jpeg, or .tiff, load it using Pillow.
- Convert to NumPy float32 array, normalized to range 0.0–1.0.
- Return an empty metadata dictionary with modality set to UNKNOWN. The modality detector (Section 4) will then attempt ML-based classification.
- For RGB images, check whether all three channels are identical. If so, convert to grayscale — most medical images are grayscale even when saved as RGB.

4. Modality Detector (core/modality_detector.py)

Correctly identifying the imaging modality is the most critical step in the pipeline. An incorrect modality assignment cascades into incorrect preprocessing, which could suppress or fabricate findings. Use a two-tier approach: fast rule-based detection from DICOM tags, with ML fallback for ambiguous or non-DICOM cases.

4.1 Tier 1 — DICOM Tag Based Detection

Read the Modality DICOM tag (0008,0060). Map standard DICOM modality codes to internal pipeline modality names as follows:

DICOM Code	Internal Name	Notes
CT	CT	Computed Tomography. Always expect Hounsfield Unit calibration.
MR	MRI	Magnetic Resonance. Must check series description for T1/T2/FLAIR/DWI subtype.
CR, DX	XRAY	CR = Computed Radiograph, DX = Digital X-ray. Both use same pipeline.
US	ULTRASOUND	Ultrasound. Speckle noise dominates.
NM	NUCLEAR_MED	Nuclear Medicine (Scintigraphy). Treat similarly to PET for preprocessing.
PT	PET	PET scan. Often co-registered with CT (PET-CT). SUV values must be preserved.
MG	MAMMOGRAPHY	Mammography. Specialized version of X-ray pipeline with micro-calcification focus.
OP, OPT	FUNDUS	Ophthalmology / Retinal imaging.
SM, XC	HISTOPATH	Slide Microscopy or External Camera capture for pathology slides.

For MRI, further sub-classify by reading the SeriesDescription tag. Look for keywords: T1 or T1W or T1-weighted maps to MRI_T1; T2 or T2W maps to MRI_T2; FLAIR maps to MRI_FLAIR; DWI or ADC maps to MRI_DIFFUSION; SWI or GRE maps to MRI_SWI. These subtypes affect which normalization strategy to use in Section 7.

4.2 Tier 2 — ML-Based Fallback

- Activate only when the Modality tag is absent, empty, or returns an unrecognized code.
- Use a pretrained lightweight image classifier. MobileNetV2 pretrained on RadImageNet is the recommended backbone. RadImageNet is a medical-domain ImageNet alternative — it gives significantly better transfer for medical modality classification than standard ImageNet weights.
- If RadImageNet weights are unavailable, fall back to a rule-based heuristic: measure the histogram characteristics of the image. CT images have bimodal histograms (air peak around -1000 HU and tissue peak around 50 HU). MRI images have roughly Gaussian-shaped histograms. X-rays have a wide dynamic range with a large low-intensity peak (lung fields). Ultrasound images have very high speckle variance.

- Always log a WARNING when the ML fallback is used. The confidence score of the classification must exceed 0.80 to proceed. Below 0.80, flag the image as MODALITY_UNCERTAIN and apply only universal safe operations (resize, bit-depth normalization) without modality-specific filtering.

5. Standardization Engine (core/standardizer.py)

Standardization brings all inputs into a consistent spatial and numerical representation before modality-specific filtering. It has no clinical effect on findings but ensures downstream consistency.

5.1 Spatial Standardization — Resizing

- Read target resolution from config per modality.
- For CT and MRI, use bicubic interpolation for downsampling and Lanczos for upsampling. Do NOT use nearest-neighbor interpolation — it creates aliasing artifacts that can mimic micro-nodules or lesions on high-contrast boundaries.
- For histopathology, do not resize individual tiles. Instead, verify that the tile extraction at the correct magnification level (typically 20× or 40× objective equivalent) has already been performed. If the input is a whole-slide image (WSI), tile it at the appropriate level before any further processing.
- Preserve aspect ratio when resizing. Use zero-padding on the shorter dimension. Center the image within the padded canvas. Log the original dimensions and padding amounts in the metadata for the LLM.

5.2 Bit-Depth and Dynamic Range Normalization

- Convert all pixel arrays to float32 as the working type.
- For CT: Do NOT normalize to 0–1 at this stage. Keep the data in Hounsfield Units. HU normalization happens inside the CT filter module using clinically defined window/level presets. Premature normalization destroys the quantitative meaning of HU values.
- For MRI: Do NOT apply global min-max normalization. Apply percentile-based normalization: clip values below the 1st percentile and above the 99th percentile of the non-zero voxel distribution, then rescale to 0.0–1.0. This avoids the common problem of bright MRI artifacts (fat saturation failures, RF noise spikes) pulling the entire image dark.
- For X-ray: Apply global min-max normalization to 0.0–1.0. X-ray pixel values do not have absolute physical units.
- For Ultrasound: Apply log-compression if the image has not already been log-compressed by the scanner. Most clinical ultrasound DICOM images arrive already log-compressed (stored as uint8 0–255). Check the dynamic range of the pixel array — if it spans more than 60 dB, log-compression is likely needed.

5.3 Orientation Correction

- For DICOM inputs, use the ImageOrientationPatient tag to determine the actual patient orientation. Use SimpleITK to resample the image into a standard anatomical orientation.
- Standard output orientation for LLM input: Axial slices in radiological convention — patient's right appears on the viewer's left. This is the convention used in most radiological training datasets and should match what the LLM was trained on.
- For chest X-rays: Ensure the image is in PA (posterior-anterior) orientation with lungs lateral. If the image is AP (anterior-posterior, commonly taken at bedside with portable units), note this in the metadata — the cardiac silhouette appears larger on AP views, which is clinically significant and the LLM must be told about it.
- For brain MRI: Standard orientation is with the patient's nose pointing forward (left in the image for axial slices). Coronal and sagittal reconstructions should be generated separately if requested.

6. Modality-Specific Noise Reduction (filters/)

This section describes each filter module individually. All filter modules must inherit from the abstract base class in `filters/base_filter.py`, which defines a single `apply(image, metadata, config)` method and a `get_quality_delta()` method that returns the SNR improvement in dB.

6.1 CT Filters (filters/ct_filters.py)

CT Noise Characteristics

CT noise is primarily quantum noise (Poisson-distributed) and follows a characteristic pattern where noise intensity increases as pixel values decrease (i.e., lung tissue is noisier than soft tissue or bone). Noise also varies with reconstruction kernel — edge-enhancing kernels (bone kernel, lung kernel) introduce high-frequency ringing artifacts that look like false structures at tissue interfaces.

Step 1: Hounsfield Unit Window/Level Application

This is the single most important CT preprocessing step for LLM findings extraction. Apply the appropriate HU window based on the body part being imaged. This converts the full 12-bit HU range into the 8-bit display range that highlights the tissue of interest.

Window Name	Center / Width (HU)	Clinical Use
Lung Window	Center: -600, Width: 1500	Pulmonary parenchyma, pneumothorax, emphysema, nodules
Mediastinum / Soft Tissue	Center: 40, Width: 400	Mediastinal masses, lymph nodes, pleural effusion, abdominal organs
Bone Window	Center: 400, Width: 1500	Fractures, cortical bone, sclerotic lesions, trabecular detail
Brain Window	Center: 35, Width: 80	Intracranial hemorrhage, ischemic stroke, mass effect
Subdural Window	Center: 75, Width: 200	Subdural hematoma — widens the window to catch isodense collections
Liver / Abdominal	Center: 60, Width: 150	Hepatic lesions, pancreas, spleen, renal masses
Vascular / Angio	Center: 250, Width: 500	CT Angiography, aorta, pulmonary embolism protocol

Window application formula: $\text{Output} = \text{clip}((\text{HU} - (\text{center} - \text{width}/2)) / \text{width} \times 255, 0, 255)$. This maps the HU range $[\text{center} - \text{width}/2, \text{center} + \text{width}/2]$ to the displayable 0–255 range.

If the LLM is intended to analyze multiple tissue types from a single CT scan, generate one windowed image per relevant window preset and pass all of them as separate inputs to the LLM. Do not attempt to merge windows — doing so destroys tissue contrast.

Step 2: Bilateral or Gaussian Denoising

- Apply a bilateral filter after windowing. Bilateral filtering reduces noise while preserving edges — critical in CT where sharp boundaries between air, soft tissue, and bone define pathology. Parameters: spatial sigma = 1.5, range sigma = 15.0 in the 0–255 scaled image.
- For thin-slice CT (slice thickness < 2mm), quantum noise is higher. Increase spatial sigma to 2.0.
- Do NOT apply Gaussian blur to chest CT lung window images. Gaussian blur reduces the sharpness of small nodules — the ACR Lung-RADS criteria requires accurate nodule size measurement, which blurring would compromise.

Step 3: Ring Artifact Detection and Flagging

- Ring artifacts appear as concentric circular bands centered on the isocenter in CT. They are caused by detector element malfunction or miscalibration. They are not pathology, but they can be misinterpreted as calcifications or lesion boundaries by an LLM.
- Detect ring artifacts by computing the variance profile along radial directions from the image center. A periodically elevated variance pattern indicates a ring artifact. Flag the image in the metadata dictionary with a `ring_artifact: true` key.
- Do not attempt to algorithmically remove ring artifacts in this pipeline — the correction requires sinogram-domain processing which is outside scope. Instead, log the flag clearly so the LLM receives the warning in its metadata context.

6.2 MRI Filters (filters/mri_filters.py)

MRI Noise Characteristics

MRI noise is Rician-distributed — not Gaussian. At high SNR, Rician noise approximates Gaussian, but at low SNR (small structures, thin slices, parallel imaging with high acceleration factor), the Rician distribution causes a noise floor bias that artificially elevates low-signal regions. Additionally, MRI images suffer from intensity non-uniformity (also called bias field) caused by spatial inhomogeneity of the RF transmit and receive coils. This is separate from pathological signal changes and must be corrected before any intensity-based analysis.

Step 1: N4 Bias Field Correction (Mandatory for Brain MRI)

- Use SimpleITK's `N4BiasFieldCorrectionImageFilter`. This is the clinical standard algorithm (N4ITK) used in virtually all neuroimaging research pipelines. It models the smooth low-frequency multiplicative bias field and divides it out.
- Apply to T1, T2, and FLAIR sequences. For DWI (diffusion-weighted imaging), bias correction is less critical because DWI uses ratio-based contrast, but apply it if the ADC map is being processed.
- N4 requires a brain mask as input for best results. If skull stripping has not been done, provide a coarse mask by thresholding out the background air (values near zero). A simple Otsu threshold is sufficient for the N4 mask — it does not need to be a perfect brain mask.
- Do NOT apply N4 bias correction to abdominal MRI — the algorithm can misinterpret T1 fat-suppression signal variation as bias field and overcorrect, removing diagnostically meaningful signal differences.

Step 2: Non-Local Means (NLM) Denoising

- Use `skimage.restoration.denoise_nl_means`. NLM is preferred over Gaussian for MRI because it searches for similar patches across the entire image and averages them — this preserves fine anatomical structures like cortical ribbons, hippocampal folds, and vessel walls that Gaussian blur would erase.
- Set the `h` parameter (filter strength) based on MRI field strength: $h = 0.05$ for 3T MRI, $h = 0.08$ for 1.5T MRI. Higher h = stronger denoising but more blurring.
- For T2-FLAIR sequences (used for white matter lesion detection in MS, small vessel disease, etc.), use conservative denoising ($h = 0.04$) because the hyperintense signal of FLAIR lesions is subtle and can be smoothed away if denoising is too aggressive.

Step 3: MRI-Specific Intensity Normalization by Subtype

- T1-weighted MRI: White matter should appear bright, gray matter intermediate, CSF dark. After NLM denoising, apply WhiteStripe normalization — identify the white matter intensity peak in the histogram and normalize so that peak sits at a fixed reference value. This is the clinical standard for multi-site MRI harmonization.
- T2-weighted MRI: Fluid (CSF, edema) appears bright, white matter dark. Apply percentile normalization (1st–99th percentile clipping and rescale to 0–1).
- FLAIR: Similar to T2 but CSF is suppressed. Apply the same percentile normalization as T2. The key diagnostic feature — hyperintense white matter lesions — should appear as bright spots after normalization.
- DWI: Do not normalize DWI images separately from their ADC maps. The ADC map encodes the apparent diffusion coefficient in physical units (mm^2/s). Normalize DWI and ADC together by clipping ADC values to the physiologically meaningful range ($0.0\text{--}3.0 \times 10^{-3} \text{ mm}^2/\text{s}$) and scaling to 0–1.

6.3 X-Ray Filters (filters/xray_filters.py)

X-Ray Noise Characteristics

Digital X-ray (CR/DX) noise has two dominant components: quantum mottle (Poisson noise from X-ray photon statistics, inversely proportional to dose) and structured noise from the detector grid. The signal-to-noise ratio varies across the image — the lungs (high air content, low attenuation) are noisier than the mediastinum or bone.

Step 1: CLAHE (Contrast-Limited Adaptive Histogram Equalization)

- CLAHE is the primary contrast enhancement method for digital X-rays. Unlike global histogram equalization, CLAHE divides the image into small tiles and equalizes each tile independently, then blends borders with bilinear interpolation. This prevents over-enhancement in already-bright regions (bone) while pulling up contrast in dim regions (lung bases, retrocardiac space).
- Apply CLAHE using OpenCV's `createCLAHE` function. Use clip limit 2.0 and tile grid size (8,8) as defaults from config.
- Convert image to `uint8` (0–255) before applying CLAHE — OpenCV's CLAHE implementation requires `uint8` or `uint16` input.
- For pediatric chest X-rays (infer from PatientAge DICOM tag if available), use a slightly lower clip limit (1.5) to avoid over-enhancing the thymus, which is physiologically prominent in children and should not be mistaken for a mediastinal mass.

Step 2: Wiener Denoising

- Apply `scipy.signal.wiener` for noise reduction after CLAHE. Wiener filtering is optimal for stationary Gaussian noise in the least-squares sense. Use a kernel size of (5,5).
- Apply Wiener filtering specifically to the lung zone of the image, not the entire image. The bone and mediastinum regions have sufficient contrast and adding smoothing there can blur fine osseous detail. Segment the lung zone using a simple threshold — lung fields typically have lower intensity than surrounding structures.

Step 3: Diaphragm and Clavicle Landmark Detection

- After filtering, attempt basic anatomical landmark localization. Detect the approximate position of the diaphragm (highest-intensity horizontal transition in the lower half of the image) and the lung apices. Add these approximate landmark coordinates to the metadata dictionary.
- This helps the downstream LLM by providing spatial reference — 'this opacity is in the right lower lobe' is more useful than 'there is an opacity in the lower right quadrant of the image'.

6.4 Ultrasound Filters (filters/ultrasound_filters.py)

Ultrasound Noise Characteristics — Speckle Is Not Random Noise

Speckle in ultrasound is a fundamental physical phenomenon caused by coherent interference of ultrasound waves scattered from sub-resolution scatterers. It is not random electronic noise. Speckle carries textural information about tissue microarchitecture. This is why aggressive speckle removal (as would be appropriate for electronic noise) is medically inappropriate — it can erase tissue texture cues that differentiate, for example, a fatty liver (bright, coarse texture) from a normal liver (uniform, medium echogenicity).

Step 1: Anisotropic Diffusion Filtering

- Use Perona-Malik anisotropic diffusion via the `medpy.filter.smoothing.anisotropic_diffusion` function. Anisotropic diffusion selectively smooths homogeneous regions (reducing speckle within organ parenchyma) while preserving edges at tissue interfaces (liver capsule, vessel walls, lesion boundaries).
- Parameters: $\kappa = 50$ (edge sensitivity threshold), $\gamma = 0.1$ (diffusion speed), number of iterations = 10. These are conservative defaults — increase iterations only if the image is very noisy.
- Do NOT use Gaussian or bilateral filtering as the primary denoising step for ultrasound. Both treat speckle as additive noise and will blur tissue boundaries.

Step 2: Log-Compression Check

- Most clinical ultrasound machines apply log compression before storing the DICOM image to convert the large dynamic range of raw echo amplitude into a displayable 8-bit range. Check whether the pixel array is already `uint8` (0–255). If so, log compression has already been applied by the scanner.
- If the pixel array has a larger dynamic range (`uint16` or float values spanning thousands), apply log compression manually: $\text{output} = \log(1 + \text{pixel}) / \log(1 + \text{max_pixel}) \times 255$.

Step 3: Gain Normalization

- Ultrasound images often have non-uniform brightness due to time-gain compensation (TGC) settings applied by the sonographer. Apply a column-wise mean normalization to correct for systematic brightness gradients along the scan depth axis.

6.5 Histopathology Filters (`filters/histopath_filters.py`)

Why Stain Normalization Is Critical

Histopathology slides are stained with Hematoxylin and Eosin (H&E). Hematoxylin stains nuclei blue-purple. Eosin stains cytoplasm and extracellular matrix pink. However, the exact color appearance varies enormously across different labs, staining protocols, scanner brands, and even between batches from the same lab. A malignant nucleus may appear deep purple in one image and pale lavender in another. Without normalization, color-sensitive LLMs will classify the same pathology differently based solely on staining variation.

Step 1: Color Space Conversion and Stain Separation

- Convert the RGB image to optical density (OD) space. $\text{OD} = -\log(\text{pixel} / 255 + \epsilon)$. This linearizes the Beer-Lambert law relationship between stain concentration and light absorption.
- Use the Macenko method (implemented in `staintools`) to separate the H&E stain matrix. Macenko uses singular value decomposition (SVD) on the OD space to estimate the stain basis vectors. This is the most widely validated method in computational pathology.
- Alternative: Use Vahadane NMF-based separation for better structure preservation when slides have many weak nuclei or sparse staining. Vahadane is slower but gives more consistent results on difficult cases.

Step 2: Stain Transfer to Reference

- Define a reference image from a well-stained, artifact-free slide. All other slides are normalized to match this reference's stain appearance.
- After Macenko separation, solve for the stain concentration maps of both the input and reference. Reconstruct the input image using the reference stain basis vectors. This transfers the reference color appearance while preserving the structural content of the input.

Step 3: Artifact Detection and Tile Rejection

- Before stain normalization, detect and reject uninformative tiles. Reject tiles that are predominantly white (background, no tissue — mean OD < 0.05). Reject tiles with a focus quality score below threshold (compute Laplacian variance; values below 50 indicate out-of-focus tiles).
- Reject tiles with prominent pen marks or artifacts (unusually saturated colors outside the H&E range, detectable by HSV saturation mapping).

6.6 Fundus (Retinal) Filters (filters/fundus_filters.py)

Fundus Imaging Characteristics

Color fundus photography captures the posterior segment of the eye including the optic disc, macula, fovea, and retinal vasculature. The primary diagnostic features — diabetic retinopathy micro-aneurysms, drusen in AMD, glaucomatous disc cupping, arteriovenous nicking in hypertension — require precise contrast in the green channel and uniform illumination.

Step 1: Green Channel Extraction and CLAHE

- Split the RGB fundus image into R, G, B channels. Process the green channel independently. The green channel provides maximum contrast for retinal vessels and lesions because hemoglobin has peak absorption at green wavelengths (~540nm). The red channel is often washed out (fundus has high red reflectance) and the blue channel is very noisy.
- Apply CLAHE to the green channel with clip limit 3.0. This enhances microvasculature and early micro-aneurysms.

Step 2: Illumination Normalization

- Fundus images suffer from uneven illumination — brighter in the center (optic disc region) and darker toward the periphery. Apply a morphological top-hat background estimation: compute a large structuring element (disk of radius 50 pixels) morphological opening of the green channel image, then subtract it. This removes the low-frequency illumination gradient while preserving fine vascular and lesion structures.

Step 3: Field of View Masking

- Fundus images typically have a circular field of view with black corners. Detect this circular boundary using a threshold-based contour method. Create a binary mask. Apply this mask to all downstream operations so the black corner pixels do not skew normalization statistics.

6.7 Dermoscopy Filters (filters/dermoscopy_filters.py)

Dermoscopy Characteristics

Dermoscopy images are high-resolution color photographs of skin lesions taken with a polarized or contact dermoscope. The main preprocessing challenges are hair artifacts (thin dark lines crossing the lesion), specular reflections (bright white spots from the glass plate), vignetting (dark circular border from the dermoscope lens), and scale bars or ruler artifacts.

Step 1: Hair Artifact Removal (Dull Razor Algorithm)

- Hair appears as thin, dark, elongated structures across the lesion. Detect them using a morphological black top-hat operation with a line-shaped structuring element rotated at multiple angles (0°, 45°, 90°, 135°). The black top-hat highlights dark structures that are thinner than the structuring element length.
- Create a binary hair mask from the top-hat output by thresholding. Dilate the mask slightly to fully cover hair width.
- Use OpenCV inpainting (Telea algorithm) to fill the detected hair regions by interpolating from surrounding skin pixels. The Telea method propagates texture from the boundary of the mask inward, producing natural-looking inpainted regions.

Step 2: Specular Reflection Removal

- Specular reflections appear as bright white saturated spots. Detect them by thresholding the V channel (value) in HSV space — pixels above the 98th percentile intensity are flagged as reflections. Inpaint these regions using the same Telea method as hair removal.

Step 3: Vignette Correction

- Dermoscope vignetting creates a radially symmetric darkening at image periphery. Estimate the vignette pattern by fitting a 2D Gaussian to the image illumination map (smoothed with a very large Gaussian kernel). Divide the original image by the normalized vignette map to flatten illumination.

7. Cross-Modality Artifact Handling

Some artifacts appear across multiple modalities and require shared utility functions. These should be implemented in the `base_filter` module or a shared utilities file and called from individual filter modules.

Artifact Type	Affected Modalities	Detection and Handling Strategy
Motion blur	MRI, CT, X-ray	Detect via Laplacian variance score < 80. Log flag. For MRI, attempt to detect motion ghosts in phase-encode direction. Do not attempt correction — flag and continue.
Truncation artifact	CT, MRI	Patient body extends beyond scanner FOV. Bright bands at image edges. Detect edge saturation. Flag in metadata.
Gibbs ringing	MRI	Oscillating ripples adjacent to sharp tissue boundaries (skull/brain interface). Apply Gibbs suppression using sub-voxel shift method from MRtrix or dipy library.
Metal artifact	CT, MRI	Bright streaks radiating from metal implants. Flag automatically if pixels > 3000 HU (CT) or > 5 standard deviations above mean (MRI). Do not remove — log warning for LLM.
Coil noise	MRI	High frequency noise at image periphery from receive coil dropout. Detect via local SNR map.
Zipper artifact	MRI	Horizontal or vertical line artifacts from RF interference. Detect as periodic horizontal banding in FFT of image. Flag for LLM.

8. Quality Assessment (core/quality_assessor.py)

Quality assessment runs twice: once on the raw image after loading (pre-processing baseline) and once on the final preprocessed image (post-processing). The delta between the two scores validates that preprocessing improved quality. If any metric worsens significantly, the pipeline must log a warning and optionally revert that step.

8.1 Metrics to Compute

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8.2 Quality Report Output

The quality assessor must return a structured dictionary with all metric values, pass/fail flags against thresholds, and a list of warnings. This dictionary must be attached to the metadata passed to the LLM formatter. The LLM should be told about image quality issues — a chest X-ray with poor SNR and motion blur should have this communicated in its context so the LLM can express appropriate uncertainty in its findings.

9. Sub-Agent Filter Selector (agents/filter_selector.py)

The filter selector is an intelligent routing module that decides which filters to apply and at what strength for a given input image. It receives the raw image, the extracted metadata, and the initial quality metrics, then outputs a filter configuration dictionary that the pipeline uses instead of the default config values.

9.1 Design Philosophy

Start with a rule-based implementation and document every rule clearly. Rules can later be replaced by or supplemented with a trained classifier. The priority order is: (1) patient safety — never use a configuration that could suppress findings, (2) quality improvement — optimize for SNR and CNR gain, (3) efficiency — prefer fewer filter steps when quality is already good.

9.2 Rule Set

- If SNR_pre > 30dB AND sharpness > 100: Image quality is already high. Apply only CLAHE and windowing. Skip denoising to avoid smoothing fine features.
- If SNR_pre < 15dB: Image quality is poor. Apply maximum-strength denoising. Add flag needs_radiologist_review to metadata.
- If motion artifact detected AND modality == MRI: Skip NLM denoising (it amplifies motion ghosts). Apply only bias field correction. Add motion_degraded flag.
- If modality == CT AND body_part == HEAD: Always use brain window AND subdural window. Never use lung window.
- If modality == CT AND body_part == CHEST: Generate three windows: lung, mediastinum, bone.
- If modality == XRAY AND PatientAge < 18 years (pediatric): Reduce CLAHE clip limit to 1.5. Flag as pediatric_case for LLM context.
- If modality == HISTOPATH AND focus_score < 50: Reject tile and do not pass to LLM. Log rejection.
- If modality == ULTRASOUND AND image_contains_doppler (detectable by color pixels in an otherwise grayscale image): Separate the grayscale B-mode component from the Doppler color overlay before processing. Process B-mode only with anisotropic diffusion. Preserve Doppler overlay separately.



DOPPLER ULTRASOUND NOTE

Color Doppler and Power Doppler overlays encode blood flow direction and velocity as color.

These color-encoded regions must NEVER be passed through grayscale normalization or color jitter.

Doppler color mapping is scanner-dependent (red vs blue = toward/away from probe, vendor-specific).

The LLM must be told in its metadata context that a Doppler overlay is present and should interpret colors accordingly.

9.3 Future ML Extension

Document the interface so it can be replaced by a trained model later. The selector should accept a feature vector containing the quality metrics, modality one-hot encoding, and body part encoding, and output a continuous parameter vector. The training signal would be the improvement in a downstream task metric (LLM finding accuracy on a validation set with ground truth). This can be framed as a reinforcement learning problem where the reward is quality delta.

10. LLM Output Formatter (core/llm_formatter.py)

This is the final stage of the pipeline. It packages the preprocessed image(s) and all accumulated metadata into the format expected by the downstream medical LLM. This step is architecturally separate from preprocessing because LLM input formats may change independently of the image processing logic.

10.1 Output Image Packaging

- Save the preprocessed image as a lossless PNG file. Never use JPEG compression for LLM input — JPEG introduces block artifacts that can be misinterpreted as lesions, particularly in chest X-rays and CT.
- If multiple window presets were generated for CT, save each as a separate PNG with the window name in the filename (e.g., image_lung_window.png, image_mediastinum_window.png).
- Include the original raw DICOM (or input image) alongside the preprocessed output. The LLM pipeline owner should always have access to the source-of-truth.
- Add a visual indicator: generate a side-by-side PNG comparing original and preprocessed for human QC. This is not sent to the LLM — it is a debug artifact logged locally.

10.2 Metadata Context String for LLM

Construct a natural language metadata context string that is sent alongside the image to the medical LLM. This string should communicate all clinically relevant context that is not visible in the image itself. Include:

- Modality and sub-type (e.g., 'Axial CT chest with lung window and mediastinum window provided')
- Body part and anatomical region
- Clinical acquisition context (e.g., 'AP portable chest radiograph — cardiac silhouette may appear enlarged relative to PA')
- Known artifacts flagged by the pipeline (e.g., 'Ring artifact detected in left lower quadrant', 'Metal artifact present from left hip prosthesis')
- Image quality summary (e.g., 'Pre-processing SNR: 18dB, Post-processing SNR: 26dB. Mild motion blur detected. Confidence in findings: moderate.')
- Preprocessing steps applied (e.g., 'CLAHE clip 2.0 applied, Wiener denoising applied, resized to 1024×1024')
- Modality-specific clinical caveats (see table below)

Modality	Mandatory Clinical Caveat to Include in LLM Context
CT (Chest)	State which window(s) are provided. Note if contrast-enhanced or non-contrast. Contrast CT will show enhancement patterns not visible on non-contrast.
CT (Brain)	Note if this is non-contrast — hemorrhage detection relies on non-contrast HU values. Contrast CT brain changes the appearance of infarcts (may appear bright from contrast enhancement rather than hemorrhage).
MRI (Brain)	State the sequence: T1 pre/post contrast, T2, FLAIR, DWI, SWI. Each sequence has different pathology visibility. Acute stroke is DWI-bright, chronic is FLAIR-bright.
X-Ray	State if PA or AP, upright or supine. Supine X-rays make pleural effusions appear as haziness rather than distinct blunting. Free air under diaphragm is only detectable on upright PA.
Ultrasound	Note transducer frequency if available. High-frequency (7–15 MHz) provides better resolution for superficial structures. Low-frequency (2–5 MHz) for deep structures. Note if B-mode only or Doppler included.

Histopath	State objective magnification level (10×, 20×, 40×). State staining type — H&E is standard; IHC (immunohistochemistry) stains require completely different interpretation. Reference slide used for normalization.
Mammography	State if screening or diagnostic. State if MLO or CC view. Note patient age — density categories (ACR BI-RADS A–D) affect lesion detection rates.
PET/PET-CT	State tracer used (FDG is most common). State SUV normalization method (body weight, lean body mass). Background SUV threshold for significant uptake: SUVmax > 2.5 for FDG.

11. Pipeline Orchestrator (pipeline.py)

The `pipeline.py` file is the single entry point. It chains all modules in strict order and handles errors gracefully. If any module fails, the pipeline must not silently pass a corrupted image to the LLM — it must either retry with safer default parameters or return a descriptive error with the last successful intermediate output.

11.1 Execution Order

Step	Action
Step 1	Load configuration from <code>config/modality_configs.yaml</code>
Step 2	Call <code>dicom_handler</code> : load image and extract metadata
Step 3	Call <code>modality_detector</code> : classify modality from metadata or ML fallback
Step 4	Call <code>quality_assessor</code> : compute baseline quality metrics (pre-processing snapshot)
Step 5	Call <code>filter_selector</code> : determine optimal filter configuration for this image
Step 6	Call <code>standardizer</code> : resize, orient, normalize bit depth
Step 7	Call modality-specific filter module: apply noise reduction and enhancement
Step 8	Call <code>quality_assessor</code> again: compute post-processing metrics. Compute delta.
Step 9	If quality delta is negative (preprocessing worsened image): revert to pre-filter image and log warning.
Step 10	Call <code>llm_formatter</code> : package image, metadata, quality report, and context string into output bundle.
Step 11	Write output bundle to disk. Log total processing time per image.

11.2 Error Handling Policy

- Wrap each step in a try-except block. Catch step-level failures individually.
- On DICOM load failure: return error immediately. Do not proceed.
- On modality detection failure: assign UNKNOWN modality. Apply only universal safe steps (Steps 6 and 10). Flag heavily in output metadata.
- On filter failure: skip that specific filter. Log the exception with full traceback. Do not crash the entire pipeline.
- On quality regression (Step 9): revert to the image state before that filter was applied. The quality assessor snapshot from Step 4 is used as the revert target.
- On LLM formatter failure: write the raw preprocessed image and raw metadata JSON as fallback output.

11.3 Batch Processing Mode

- The pipeline should accept a directory path as input in addition to a single file path.
- In batch mode, process images in parallel using Python's `concurrent.futures.ProcessPoolExecutor`. Set `max_workers` based on CPU count.
- Generate a batch summary report in JSON listing per-image processing time, quality metrics, modality detected, warnings flagged, and output paths.

12. Testing Strategy (tests/test_pipeline.py)

12.1 Sample Data to Acquire

Obtain at least one sample image per modality from publicly available medical imaging datasets. Use the following sources:

- CT Chest: LIDC-IDRI dataset (lung nodule CT, freely available via TCIA — The Cancer Imaging Archive)
- Brain MRI: BraTS 2023 dataset (T1, T2, FLAIR, DWI all included for brain tumor cases)
- Chest X-ray: NIH Chest X-ray14 dataset (100,000+ frontal chest PA views)
- Ultrasound: Point-of-Care Ultrasound (POCUS) dataset on PhysioNet
- Histopathology: CAMELYON17 dataset (whole-slide images for breast cancer lymph node metastasis)
- Fundus: DRIVE dataset (retinal vessel segmentation, 40 color fundus images)

12.2 Tests to Write

- Test that MONOCHROME1 inversion is correctly applied when PhotometricInterpretation tag equals MONOCHROME1.
- Test that CT HU window application produces expected pixel value distributions: lung window should have very few pixels in the 0–100 HU range (mostly air and lung), mediastinum window should cluster near 40–80 HU.
- Test that MRI N4 bias field correction reduces the coefficient of variation of white matter intensity (a standard validation metric for bias correction quality).
- Test that the quality delta is positive (preprocessing improved quality) for all standard sample inputs.
- Test that histopathology blank tile rejection correctly filters out tiles with mean OD below 0.05.
- Test that the pipeline does not crash and produces a valid output bundle for every modality sample.
- Test the LLM formatter context string for each modality — verify that the mandatory clinical caveats from Section 10.2 are present in the output string.

13. Common Mistakes to Avoid

DO NOT DO THESE

Do NOT apply CLAHE to CT images. CT uses window/level for contrast adjustment, not histogram equalization. CLAHE would destroy Hounsfield Unit meaning.

Do NOT apply NLM denoising before bias field correction on MRI. The bias field adds spatially varying intensity that NLM will try to preserve as 'signal', making bias correction less effective afterward.

Do NOT globally normalize CT images to 0–1 before windowing. Apply windowing first, then normalize the windowed image. Global normalization before windowing will compress the diagnostically useful range.

Do NOT use JPEG output for any processed medical image. JPEG compression creates block artifacts that resemble calcifications, reticulation, and ground-glass opacity.

Do NOT apply color augmentation (hue shift, saturation, brightness) to CT, MRI, PET, or nuclear medicine images. These modalities use quantitative pixel values that encode physical tissue properties.

Do NOT use nearest-neighbor interpolation for resizing. It creates aliasing artifacts that mimic micro-nodules and small calcifications.

Do NOT treat Doppler color overlays as regular image content. Separate them before grayscale processing.

Do NOT skip the PhotometricInterpretation check. A missed MONOCHROME1 inversion will cause the LLM to analyze an inverted image — bones appear dark, lungs appear white.

ALWAYS DO THESE

Always preserve the original DICOM file unchanged. Write processed outputs to a separate directory.

Always log the modality detection method used (DICOM tag vs ML fallback) and confidence score.

Always include quality metrics in the LLM context string. The LLM must know if an image has quality limitations.

Always handle multi-frame DICOM (3D volumes) — most real CT and MRI data is volumetric.

Always check for and handle the MONOCHROME1 photometric interpretation.

Always apply HU Rescale Slope and Intercept correction for CT before any other processing.

Always use lossless PNG output for the LLM-ready image.

Always generate a side-by-side QC visualization for human review.

End of Implementation Guide — v1.0